

Operations on sets



- 1
 - a The set of all the elements in the universal set that are not in set A .
 - b The set containing all the elements that are members of set A or of set B or of both sets.
 - c The set containing all the elements that are common to both set A and set B .
 - d The set of elements that belong to set A , but not to set B .

- 2 $\{3, 7\}$

- 3
 - a $\{0, 4, 16, 36, 64\}$
 - b $\{1, 3, 5, 7, 15, 21, 35, 105\}$
 - c $\{5, 10, 15, 20, 25, 30, 35, 40, 45\}$

- 4
 - a $\{1, 2, 3, 4, 5, 6\}$
 - b $\{1, 2, 3, 4, 5, 6, 7, 8\}$
 - c $\{1, 2, 3, 4, 5, 6\}$
 - d $\{1, 2, 3, 4, 5, 6, 7, 8\}$

- 5
 - a
 - i $\{1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36\}$
 - ii $\{1, 2, 3, 4, 6, 12\}$
 - b
 - i $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 17, 19, 23, 29, 31, 37\}$
 - ii $\{2, 3, 5, 7, 11\}$

- 6
 - a
 - i The universal set.
 - ii The universal set, except for the elements of A and the elements of B .
 - b
 - i The universal set.
 - ii The universal set, except for the elements of A and the elements of B .

- 7
 - a The universal set, except for the elements that are in both A and B .
 - b The set of common multiples of 4 and 6 greater than or equal to 70.
 - c No



- 8
- a The set of integers that are not multiples of 4 less than 70 or multiples of 6 less than 70.
 - b The set of integers that are not less than 70, and are multiples of both 4 and 6.
 - c No

- 9
- a $\{n, p, m\}$
 - b $\{f, a\}$

- 10
- a $\{85, 20\}$
 - b $\{19, 69, 17, 35, 29, 85, 20\}$

Complementary events

- 11
- a Not complementary
 - b Complementary
 - c Not complementary
 - d Not complementary

- 12
- a $\frac{1}{5}$
 - b 0.36

- 13 0.44

- 14
- a $\frac{34}{69}$
 - b $\frac{35}{69}$

- 15
- a $\frac{37}{139}$
 - b $\frac{60}{139}$
 - c $\frac{102}{139}$
 - d $\frac{79}{139}$

- 16
- a Picking a number less than or equal to 61.
 - b $\frac{39}{100}$

17

a $\frac{5}{6}$
e 0

b $\frac{2}{3}$



c $\frac{1}{2}$

d 1

18

a $\frac{25}{26}$
e $\frac{21}{26}$

b $\frac{23}{26}$

c $\frac{17}{26}$

d $\frac{10}{13}$

19

a $\frac{1}{4}$

b $\frac{1}{4}$

c $\frac{3}{4}$

20

a $\frac{1}{13}$
e $\frac{1}{2}$

b $\frac{1}{13}$

c $\frac{12}{13}$

d $\frac{1}{2}$

21

a 66

b 20

22

a $\frac{19}{30}$

b $\frac{9}{10}$

23

a $\frac{3}{13}$
e $\frac{25}{26}$
i $\frac{12}{13}$

b $\frac{1}{26}$
f $\frac{51}{52}$

c $\frac{2}{13}$
g $\frac{5}{13}$

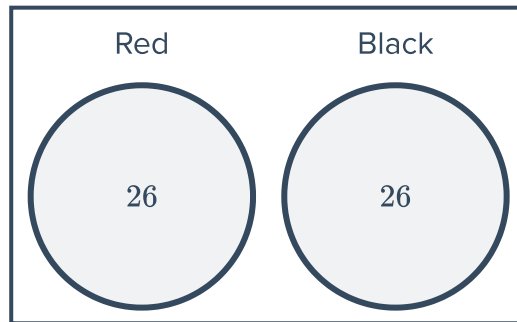
d $\frac{1}{26}$
h $\frac{1}{52}$

24 $\frac{5}{36}$



Probabilities from Venn diagrams

25



26

a $\frac{30}{67}$
e $\frac{19}{67}$

b $\frac{32}{67}$

c $\frac{37}{67}$

d $\frac{35}{67}$

27

a $\frac{2}{9}$

b $\frac{7}{54}$

c $\frac{19}{27}$

d $\frac{8}{27}$

28

a i Yes ii No iii No iv No

b $\frac{9}{20}$

29

a $\frac{16}{33}$

b $\frac{7}{33}$

c $\frac{38}{99}$

d $\frac{89}{99}$

30

a $\frac{145}{224}$
e $\frac{3}{112}$

b 0

c $\frac{73}{224}$

d $\frac{109}{112}$

31

a $\frac{314}{535}$
e $\frac{36}{535}$

b $\frac{246}{535}$
f $\frac{37}{107}$

c $\frac{68}{535}$

d $\frac{499}{535}$

32

a

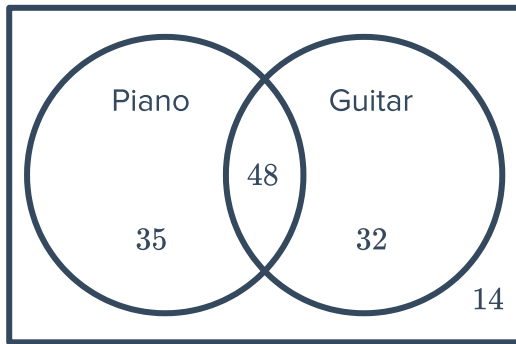


b

i $\frac{16}{43}$

ii $\frac{115}{129}$

iii $\frac{14}{129}$



33

a 18

b $\frac{93}{172}$

c $\frac{93}{172}$

34

a $\frac{565}{1232}$

b $\frac{235}{1232}$

c $\frac{923}{1232}$

d $\frac{131}{1232}$

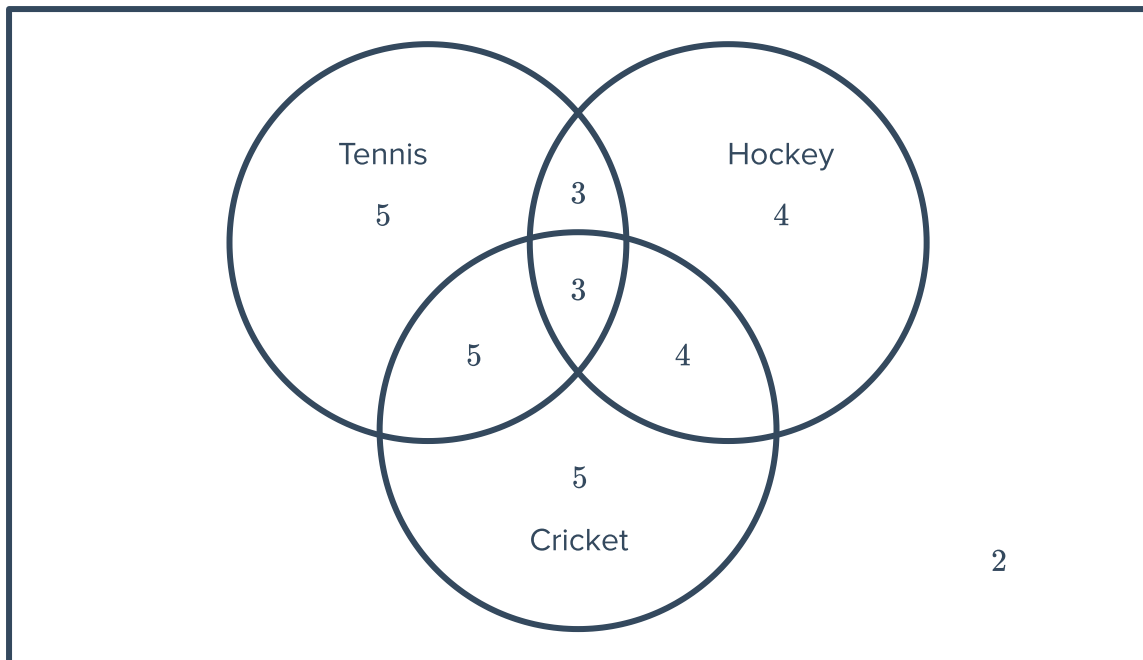
e $\frac{13}{154}$

f $\frac{93}{616}$

g $\frac{159}{176}$

35

a



b $\frac{3}{31}$

